

**General instructions:** After reading the questions given below, follow the steps described in `Instructions.pdf` to see the input and the expected output for each test case.

## 1 Can you Connect the Dots?!

You are given an array `points` representing integer coordinates of some points on a 2D-plane, where `points[i] = [xi, yi]`.

The cost of connecting two points  $[x_i, y_i]$  and  $[x_j, y_j]$  is the Manhattan distance between them:  $|x_i - x_j| + |y_i - y_j|$ , where  $|val|$  denotes the absolute value of  $val$ .

Return the minimum cost to make all points connected. All points are connected if there is exactly one simple path between any two points.

### Input Format:

- The first line of input contains a single integer  $t$  - the number of test cases. Then the descriptions of the test cases follow.
- The first line of each test case contains a single integer  $n$  - the size of array `points`.
- $n$  lines follow wherein each line contains two integers  $x$  and  $y$ . For the  $i_{th}$  line,  $x$  and  $y$  correspond to  $[x_i, y_i]$  such that `points[i] = [xi, yi]`.

### Output Format:

For each test case, print a single integer - the minimum cost to make all points connected.

### Constraints:

$$1 \leq t \leq 1000$$

$$1 \leq n \leq 1000$$

$$-10^6 \leq x_i, y_i \leq 10^6$$

### Expected Time Complexity:

$$O(n^2)$$

**TEST CASE 1****Input:**

2  
5  
0 0  
2 2  
3 10  
5 2  
7 0  
3  
3 12  
-2 5  
-4 1

**Output:**

20  
18

**Explanation:**

*1st Test Case:*

Connect the following points to get the minimum cost:

1. [0,0] and [2,2]
2. [2,2] and [3,10]
3. [2,2] and [5,2]
4. [5,2] and [7,0]

*2nd Test Case:*

Connect the following points to get the minimum cost:

1. [-4,1] and [-2,5]
2. [-2,5] and [3,12]

## 2 Discount Coupon

Your task is to find a minimum-price flight route from City 1 to City  $n$  in a graph network of  $n$  cities. You have **one** discount coupon, using which you can halve the price of any single non-stop flight during the route. However, you can use the coupon **only once**.

**Note:** When you use the discount coupon for a flight whose price is  $x$ , its price becomes  $\lfloor x/2 \rfloor$  (it is rounded down to an integer).

**Input Format:**

- The first input line has two integers  $n$  and  $m$ : the number of cities and flight connections. The cities are numbered  $1, 2, \dots, n$ .
- After this there are  $m$  lines describing the flights. Each line has three integers  $a$ ,  $b$ , and  $c$ : a flight begins at city  $a$ , ends at city  $b$ , and its price is  $c$ . Each flight is **unidirectional**.
- You can assume that it is always possible to get from City 1 to City  $n$ .

**Output Format:**

Return one integer: the price of the cheapest route from City 1 to City  $n$  with the provision of using one discount coupon.

**Constraints:**

$$2 \leq n \leq 10^5$$

$$1 \leq m \leq 2 \cdot 10^5$$

$$1 \leq a, b \leq n$$

$$1 \leq c \leq 10^9$$

**Expected Time Complexity:**  $O((n + m)\log(m))$

**TEST CASE 1****Input:**

3 4

1 2 3

2 3 1

1 3 7

2 1 5

**Output:**

2

**Explanation:**

The shortest path with discount is from  $1 \rightarrow 2 \rightarrow 3$  with cost =  $\lfloor 3 \rfloor + 1 = 1 + 1 = 2$

**TEST CASE 2****Input:**

3 3

1 2 3

2 3 3

1 3 7

**Output:**

3

**Explanation:**

The shortest path with discount is from  $1 \rightarrow 3$  with cost  $= \lfloor 7 \rfloor = 3$ . Note this is **different** from the shortest path  $1 \rightarrow 2 \rightarrow 3$  where with discount the cost would be  $= \lfloor 3 \rfloor + 3 = 4 > 3$